

Mikhail Razakov

Compiler & Program Analysis Engineer

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Compiler and program-analysis engineer. I work on C#.NET static analysis at ISP RAS and previously worked on LLVM 22/C++ compiler engineering at MCST. Public work covers interprocedural LLVM analysis with explicit legality checks, Roslyn CFG fixed-point data flow, and an x86-64 backend.

PROFESSIONAL EXPERIENCE

2026 — present ISP RAS — Static Analysis Engineer

- Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform.
- Current work includes symbolic-computation tasks in C#.NET program analysis.

July — August 2026 MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves; a separate public native-codegen lineage starts with Visk (2023) and Wist4 (2024).

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

SKILLS

Compiler / IR

C++23, LLVM 22/IR, APInt, interprocedural analysis, CFG/SSA, dominators, loops, LICM, legality

Program Analysis

C#.NET, Roslyn ControlFlowGraph, fixed-point data flow, conditional edges, loops/back edges, joins

Backend / Codegen

x86-64, SysV ABI, NASM IA-32, CDECL/x87, liveness/interference, register allocation, spills, phi lowering

Correctness / Tooling

LLVM Verifier, conservative rejection, regression checks, differential execution, Linux, CMake

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhi Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote